

NCERT Solutions for Class 12

Maths

Chapter 2 Inverse Trigonometric Functions

Exercise 2.1

1. The principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is

Ans: Assuming $\sin^{-1}\left(-\frac{1}{2}\right)=x$

Further solving,

$$\begin{aligned}\sin x &= \left(-\frac{1}{2}\right) \\ &= \sin\left(-\frac{\pi}{6}\right)\end{aligned}$$

The principal value of $\sin^{-1}x$ lies in the range of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Therefore, the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is $-\frac{\pi}{6}$

2. The principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is

Ans: Assuming $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)=x$

Further solving,

$$\begin{aligned}\cos x &= \left(\frac{\sqrt{3}}{2}\right) \\ &= \cos\left(\frac{\pi}{6}\right)\end{aligned}$$

The principal value of $\cos^{-1}x$ lies in the range of $[0, \pi]$

Therefore, the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $\frac{\pi}{6}$

3. The principal value of $\operatorname{cosec}^{-1}(2)$ is

Ans: Assuming $\operatorname{cosec}^{-1}(2)=x$

Further solving,

$$\operatorname{cosec}(x)=2$$

$$=\operatorname{cosec}\left(\frac{\pi}{6}\right)$$

The principal value of $\operatorname{cosec}^{-1}x$ lies in the range of $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]-\{0\}$

Therefore, the principal value of $\operatorname{cosec}^{-1}(2)$ is $\frac{\pi}{6}$

4. The principal value of $\tan^{-1}(-\sqrt{3})$ is

Ans: Assuming $\tan^{-1}(-\sqrt{3})=x$

Further solving,

$$\tan x=-\sqrt{3}$$

$$=\tan\left(-\frac{\pi}{3}\right)$$

The principal value of $\tan^{-1}x$ lies in the range of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Therefore, the principal value of $\tan^{-1}(-\sqrt{3})$ is $-\frac{\pi}{3}$

5. The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is

Ans: Assuming $\cos^{-1}\left(-\frac{1}{2}\right)=x$

Further solving,

$$\cos x=-\frac{1}{2}$$

$$=\cos\left(\frac{2\pi}{3}\right)$$

The principal value of $\cos^{-1}x$ lies in the range of $[0, \pi]$

Therefore, the principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is $\frac{2\pi}{3}$

6. The principal value of $\tan^{-1}(-1)$ is

Ans: Assuming $\tan^{-1}(-1)=x$

Further solving,

$$\begin{aligned}\tan x &= -1 \\ &= \tan\left(-\frac{\pi}{4}\right)\end{aligned}$$

The principal value of $\tan^{-1}x$ lies in the range of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Therefore, the principal value of $\tan^{-1}(-1)$ is $-\frac{\pi}{4}$

7. The principal value of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ is

Ans: Assuming $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)=x$

Further solving,

$$\begin{aligned}\sec x &= \frac{2}{\sqrt{3}} \\ &= \sec\left(\frac{\pi}{6}\right)\end{aligned}$$

The principal value of $\sec^{-1}x$ lies in the range of $[0, \pi]$

Therefore, the principal value of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ is $\frac{\pi}{6}$

8. The principal value of $\cot^{-1}(\sqrt{3})$ is

Ans: Assuming $\cot^{-1}(\sqrt{3})=x$

Further solving,

$$\begin{aligned}\cot x &= \sqrt{3} \\ &= \cot\left(\frac{\pi}{6}\right)\end{aligned}$$

The principal value of $\cot^{-1}x$ lies in the range of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Therefore, the principal value of $\cot^{-1}(\sqrt{3})$ is $\frac{\pi}{6}$

9. The principal value of $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ is

Ans: Assuming $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)=x$

Further solving,

$$\begin{aligned}\cos x &= -\frac{1}{\sqrt{2}} \\ &= \cos\left(\frac{3\pi}{4}\right)\end{aligned}$$

The principal value of $\cos^{-1}x$ lies in the range of $[0, \pi]$

Therefore, the principal value of $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ is $\frac{3\pi}{4}$

10. The principal value of $\operatorname{cosec}^{-1}(-\sqrt{2})$ is

Ans: Assuming $\operatorname{cosec}^{-1}(-\sqrt{2})=x$

Further solving,

$$\begin{aligned}\operatorname{cosec} x &= -\sqrt{2} \\ &= \operatorname{cosec}\left(-\frac{\pi}{4}\right)\end{aligned}$$

The principal value of $\operatorname{cosec}^{-1}x$ lies in the range of $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Therefore, the principal value of $\operatorname{cosec}^{-1}(-\sqrt{2})$ is $-\frac{\pi}{4}$

11. Evaluate $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

Ans: Assuming that $\tan^{-1}(1)=a$

$$\tan a = 1$$

$$= \tan \frac{\pi}{4}$$

$$\tan^{-1}(1) = \frac{\pi}{4} \text{ -----(1)}$$

Assuming that $\cos^{-1}\left(\frac{1}{2}\right) = b$

$$\cos b = \frac{1}{2}$$

$$= \cos \frac{\pi}{3}$$

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ -----(2)}$$

Assuming that $\sin^{-1}\left(\frac{1}{2}\right) = c$

$$\sin c = \frac{1}{2}$$

$$= \sin \frac{\pi}{6}$$

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \text{ -----(3)}$$

From Equations (1), (2) and (3),

$$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$$

$$= \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6}$$

$$= \frac{3\pi}{4}$$

Therefore, $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{3\pi}{4}$

12. Evaluate $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$

Ans: Assuming that $\cos^{-1}\left(\frac{1}{2}\right) = a$

$$\cos a = \frac{1}{2}$$

$$= \cos \frac{\pi}{3}$$

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ -----(1)}$$

$$\text{Assuming that } 2\sin^{-1}\left(\frac{1}{2}\right) = b$$

$$\sin b = \frac{1}{2}$$

$$= \sin \frac{\pi}{6}$$

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$2\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ -----(2)}$$

From Equations (1) and (2),

$$\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3} + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

$$\text{Therefore, } \cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right) = \frac{2\pi}{3}$$

13. Range of y if $\sin^{-1}x=y$ is

(A) $0 \leq y \leq \pi$

(B) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(C) $0 < y < \pi$

(D) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

Ans: Given that $\sin^{-1}x=y$

Since, the range of $\sin^{-1}x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Therefore the range of y is $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

14. Evaluate $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$

Ans: Assuming that $\tan^{-1}(\sqrt{3}) = a$

$$\tan a = \sqrt{3}$$

$$= \tan \frac{\pi}{3}$$

$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \text{ -----(1)}$$

Assuming that $\sec^{-1}(-2) = b$

$$\sec b = -2$$

$$= \sec\left(\frac{2\pi}{3}\right)$$

$$\sec^{-1}(-2) = \frac{2\pi}{3} \text{ -----(2)}$$

From Equations (1) and (2),

$$\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$$

$$= \frac{\pi}{3} - \frac{2\pi}{3}$$

$$= -\frac{\pi}{3}$$

Therefore, $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2) = -\frac{\pi}{3}$

Exercise 2.2

1. Using inverse trigonometric identities, prove that

$$3\sin^{-1}x = \sin^{-1}(3x - 4x^3), x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Ans: Assuming $\sin^{-1}x = \alpha$

$$\sin \alpha = x$$

We know that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$

Consider

$$\begin{aligned}
 & \sin^{-1}(3x-4x^3) \\
 &= \sin^{-1}(3\sin\alpha-4\sin^3\alpha) \\
 &= \sin^{-1}(\sin 3\alpha) \\
 &= 3\alpha \\
 &= 3\sin^{-1}x \\
 & \text{Hence it is proved that } 3\sin^{-1}x = \sin^{-1}(3x-4x^3)
 \end{aligned}$$

2. Using inverse trigonometric identities, prove that

$$3\cos^{-1}x = \cos^{-1}(4x^3 - 3x), x \in \left[\frac{1}{2}, 1\right]$$

Ans: Assuming $\cos^{-1}x = \alpha$

$$\cos\alpha = x$$

We know that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

Consider

$$\begin{aligned}
 & \cos^{-1}(4x^3 - 3x) \\
 &= \cos^{-1}(4\cos^3\alpha - 3\cos\alpha) \\
 &= \cos^{-1}(\cos 3\alpha) \\
 &= 3\alpha \\
 &= 3\cos^{-1}x
 \end{aligned}$$

Hence it is proved that $3\cos^{-1}x = \cos^{-1}(4x^3 - 3x)$

3. Using inverse trigonometric identities, prove that $\tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} = \tan^{-1}\frac{1}{2}$

Ans: We know that $\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Consider

$$\begin{aligned}
 & \tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} \\
 &= \tan^{-1}\frac{\frac{2}{11} + \frac{7}{24}}{1 - \frac{2}{11} \cdot \frac{7}{24}} \\
 &= \tan^{-1}\left(\frac{1}{2}\right)
 \end{aligned}$$

Hence it is proved that $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \left(\frac{1}{2} \right)$

4. Using inverse trigonometric identities, prove that $2\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

Ans: We know that $\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$

Consider

$$\begin{aligned} & 2\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} \\ &= \tan^{-1} \frac{4}{3} + \tan^{-1} \frac{1}{7} \\ &= \tan^{-1} \left(\frac{\frac{4}{3} + \frac{1}{7}}{1 - \left(\frac{4}{3} \times \frac{1}{7} \right)} \right) \\ &= \tan^{-1} \left(\frac{31}{17} \right) \end{aligned}$$

Hence it is proved that $2\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

5. Simplify $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}, x \neq 0$

Ans: Consider $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$

Assuming

$$x = \tan \theta$$

$$\beta = \tan^{-1} x$$

$$\begin{aligned} & \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right) \\ &= \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \beta}-1}{\tan \beta} \right) \end{aligned}$$

$$= \tan^{-1} \left(\frac{\sec \beta - 1}{\tan \beta} \right)$$

$$= \tan^{-1} \left(\frac{1 - \cos \beta}{\sin \beta} \right)$$

$$= \tan^{-1} \left(\tan \left(\frac{\beta}{2} \right) \right)$$

$$= \frac{\beta}{2}$$

$$= \frac{1}{2} \tan^{-1} x$$

Therefore, $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x} = \frac{1}{2} \tan^{-1} x$

6. Simplify $\tan^{-1} \frac{1}{\sqrt{x^2-1}}$, $|x| > 1$

Ans: Consider $\tan^{-1} \frac{1}{\sqrt{x^2-1}}$

Assuming
 $x = \sec \beta$

$\beta = \sec^{-1} x$

$$\tan^{-1} \frac{1}{\sqrt{x^2-1}}$$

$$= \tan^{-1} \left(\frac{1}{\sqrt{\sec^2 \beta - 1}} \right)$$

$$= \tan^{-1} \left(\frac{1}{\tan \beta} \right)$$

$$= \tan^{-1} (\cot \beta)$$

$$= \frac{\pi}{2} - \beta$$

$$= \frac{\pi}{2} - \sec^{-1} x$$

Therefore, $\tan^{-1} \frac{1}{\sqrt{x^2-1}} = \frac{\pi}{2} - \sec^{-1} x$

7. Simplify $\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right), x < \pi$

Ans: Consider $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$

$$\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$$

$$= \tan^{-1}\sqrt{\frac{2\sin^2\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)}}$$

$$= \tan^{-1}\left(\tan\left(\frac{x}{2}\right)\right)$$

$$= \frac{x}{2}$$

Therefore, $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{x}{2}$

8. Simplify $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right), 0 < x < \pi$

Ans: Consider $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$

$$\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$$

$$= \tan^{-1}\left(\frac{1 - \tan x}{1 + \tan x}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - x\right)\right)$$

$$= \frac{\pi}{4} - x$$

Therefore, $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right) = \frac{\pi}{4} - x$

9. Simplify $\tan^{-1} \frac{x}{\sqrt{a^2-x^2}}, |x| < a$

Ans: Consider $\tan^{-1} \frac{x}{\sqrt{a^2-x^2}}$

Assuming

$$x = a \sin \alpha$$

$$\tan^{-1} \frac{x}{\sqrt{a^2-x^2}}$$

$$= \tan^{-1} \left(\frac{a \sin \alpha}{\sqrt{a^2 - a^2 \sin^2 \alpha}} \right)$$

$$= \tan^{-1} (\tan \alpha)$$

$$= \alpha$$

$$= \sin^{-1} \left(\frac{x}{a} \right)$$

$$\text{Therefore, } \tan^{-1} \frac{x}{\sqrt{a^2-x^2}} = \sin^{-1} \left(\frac{x}{a} \right)$$

10. Simplify $\tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right), a > 0; \frac{-a}{\sqrt{3}} \leq x \leq \frac{a}{\sqrt{3}}$

Ans: Consider $\tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right)$

Assuming

$$x = a \tan \alpha$$

$$\tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right)$$

$$= \tan^{-1} \left(\frac{3a^3 \tan \alpha - a^3 \tan^3 \alpha}{a^3 - 3a^3 \tan^2 \alpha} \right)$$

$$= \tan^{-1} (\tan 3\alpha)$$

$$= 3\alpha$$

$$= 3 \tan^{-1} \left(\frac{x}{a} \right)$$

$$\text{Therefore, } \tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right) = 3 \tan^{-1} \left(\frac{x}{a} \right)$$

11. Evaluate $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$

Ans: Assuming $\sin^{-1}\frac{1}{2}=a$

$$\sin a = \frac{1}{2}$$

$$= \sin\left(\frac{\pi}{6}\right)$$

$$\sin^{-1}\frac{1}{2} = \frac{\pi}{6} \text{ -----(1)}$$

Further solving,

From equation (1)

$$\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$$

$$= \tan^{-1}\left[2\cos\left(2 \times \frac{\pi}{6}\right)\right]$$

$$= \tan^{-1}\left[2\cos\frac{\pi}{3}\right]$$

$$= \tan^{-1}(1)$$

$$= \frac{\pi}{4}$$

$$\text{Therefore, } \tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right] = \frac{\pi}{4}$$

12. Evaluate $\cot(\tan^{-1}a + \cot^{-1}a)$

Ans: Consider

$$\tan^{-1}a + \cot^{-1}a$$

$$= \frac{\pi}{2}$$

Further solving,

$$\cot(\tan^{-1}a + \cot^{-1}a)$$

$$= \cot\left(\frac{\pi}{2}\right)$$

$$= 0$$

Therefore, $\cot(\tan^{-1}a + \cot^{-1}a) = 0$

13. Evaluate $\tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1-x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], |x| < 1, y > 0 \text{ and } xy < 1$

Ans: Assuming $x = \tan \alpha$

Consider

$$\begin{aligned} & \sin^{-1} \left(\frac{2x}{1-x^2} \right) \\ &= \sin^{-1} \left(\frac{2 \tan \alpha}{1 - \tan^2 \alpha} \right) \\ &= \sin^{-1} (\sin 2\alpha) \\ &= 2\alpha \\ &= 2 \tan^{-1} x \text{ -----(1)} \end{aligned}$$

Consider $\cos^{-1} \left(\frac{1-y^2}{1+y^2} \right)$

Assuming $y = \tan \beta$

$$\begin{aligned} & \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \\ &= \cos^{-1} \left(\frac{1 - \tan^2 \beta}{1 + \tan^2 \beta} \right) \\ &= \cos^{-1} (\cos 2\beta) \\ &= 2\beta \\ &= 2 \tan^{-1} y \text{ -----(2)} \end{aligned}$$

From equations (1) and (2),

$$\begin{aligned} & \tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1-x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right] \\ &= \tan \frac{1}{2} [2 \tan^{-1} x + 2 \tan^{-1} y] \\ &= \tan (\tan^{-1} x + \tan^{-1} y) \\ &= \frac{x+y}{1-xy} \end{aligned}$$

Therefore, $\tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1-x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right] = \frac{x+y}{1-xy}$

14. Compute the value of x which satisfies $\sin \left(\sin^{-1} \frac{1}{5} + \cos^{-1} x \right) = 1$

Ans: We know that $\sin(A+B) = \sin A \cos B + \cos A \sin B$

Consider $\sin \left(\sin^{-1} \frac{1}{5} + \cos^{-1} x \right) = 1$

$$\sin \left(\sin^{-1} \frac{1}{5} \right) \cos(\cos^{-1} x) + \cos \left(\sin^{-1} \frac{1}{5} \right) \sin(\cos^{-1} x) = 1$$

$$\frac{x}{5} + \cos \left(\sin^{-1} \frac{1}{5} \right) \sin(\cos^{-1} x) = 1 \text{ ----(1)}$$

Assuming

$$\sin^{-1} \frac{1}{5} = a$$

$$\sin a = \frac{1}{5}$$

$$\cos a = \sqrt{1 - \left(\frac{1}{5} \right)^2}$$

$$\cos a = \frac{2\sqrt{6}}{5}$$

$$a = \cos^{-1} \left(\frac{2\sqrt{6}}{5} \right)$$

$$\sin^{-1} \left(\frac{1}{5} \right) = \cos^{-1} \left(\frac{2\sqrt{6}}{5} \right) \text{ ----(2)}$$

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Assuming

$$\cos^{-1} x = b$$

$$\cos b = x$$

$$\sin b = \sqrt{1-x^2}$$

$$b = \sin^{-1} \left(\sqrt{1-x^2} \right)$$

$$\cos^{-1}x = \sin^{-1}(\sqrt{1-x^2}) \text{ -----(3)}$$

From Equations (1), (2) and (3),

$$\frac{x}{5} + \frac{2\sqrt{6}}{5}\sqrt{1-x^2} = 1$$

$$x + 2\sqrt{6-6x^2} = 5$$

$$(5x-1)^2 = 0$$

$$x = \frac{1}{5}$$

Therefore, for $x = \frac{1}{5}$, $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$

15. Compute the value of x which satisfies $\tan^{-1}\frac{x-1}{x-2} + \tan^{-1}\frac{x+1}{x+2} = \frac{\pi}{4}$

Ans: We know that $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Consider

$$\tan^{-1}\frac{x-1}{x-2} + \tan^{-1}\frac{x+1}{x+2} = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \left(\frac{x-1}{x-2} \times \frac{x+1}{x+2}\right)}\right) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{2x^2-4}{-3}\right) = \frac{\pi}{4}$$

$$\frac{2x^2-4}{-3} = 1$$

$$x = \pm \frac{1}{\sqrt{2}}$$

Therefore, for $x = \pm \frac{1}{\sqrt{2}}$, $\tan^{-1}\frac{x-1}{x-2} + \tan^{-1}\frac{x+1}{x+2} = \frac{\pi}{4}$

16. Evaluate $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

Ans: We know that $\sin(\theta) = \sin(\pi - \theta)$

Consider $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$

$$\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$$

$$=\sin^{-1}\left(\sin\left(\pi-\frac{2\pi}{3}\right)\right)$$

$$=\sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$$

$$=\frac{\pi}{3}$$

Therefore, $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)=\frac{\pi}{3}$

17. Evaluate $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$

Ans: We know that $\tan(\theta)=-\tan(-\theta)$

Consider $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$

$$\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$$

$$=\tan^{-1}\left(-\tan\left(-\frac{3\pi}{4}\right)\right)$$

$$=\tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right)$$

$$=-\frac{\pi}{4}$$

Therefore, $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)=\frac{\pi}{4}$

18. Evaluate $\tan\left(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2}\right)$

Ans: Assuming $\sin^{-1}\frac{3}{5}=a$

$$\sin^{-1} \frac{3}{5} = a$$

$$\sin a = \frac{3}{5}$$

$$\cos a = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$\cos a = \frac{4}{5}$$

$$\tan a = \frac{3}{4}$$

$$a = \tan^{-1} \left(\frac{3}{4} \right)$$

$$\sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \text{ ----(1)}$$

$$\cot^{-1} \frac{3}{2} = \tan^{-1} \frac{2}{3} \text{ -----(2)}$$

Further solving,

$$\tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$$

$$= \tan \left(\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right)$$

$$= \tan \left(\tan^{-1} \frac{\frac{3}{4} + \frac{2}{3}}{1 - \left(\frac{3}{4} \times \frac{2}{3} \right)} \right)$$

$$= \tan \left(\tan^{-1} \frac{17}{6} \right)$$

$$= \frac{17}{6}$$

$$\text{Therefore, } \tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right) = \frac{17}{6}$$

19. Value of $\cos\left(\cos^{-1}\frac{7\pi}{6}\right)$ is

- (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Ans: Consider $\cos\left(\cos^{-1}\frac{7\pi}{6}\right)$

$$=\cos\left(\cos^{-1}\frac{-7\pi}{6}\right)$$

$$=\cos^{-1}\left[\cos\frac{5\pi}{6}\right]$$

$$=\frac{5\pi}{6}$$

$$\text{Therefore, } \cos\left(\cos^{-1}\frac{7\pi}{6}\right)=\frac{5\pi}{6}$$

The correct option is B

20. Value of $\sin\left(\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right)$ is

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

Ans: Assuming $\sin^{-1}\left(-\frac{1}{2}\right)=a$

$$\sin a = -\frac{1}{2}$$

$$=-\sin\left(\frac{\pi}{6}\right)$$

$$=\sin\left(-\frac{\pi}{6}\right)$$

$$\sin^{-1}\left(-\frac{1}{2}\right)=-\frac{\pi}{6}$$

Further solving,

$$\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$$

$$= \sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right)$$

$$= \sin\left(\frac{\pi}{2}\right)$$

$$= 1$$

$$\text{Therefore, } \sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right) = 1$$

The correct option is D

Miscellaneous Solutions

1. Evaluate $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

Ans: Consider $\cos\frac{13\pi}{6}$

$$\cos\frac{13\pi}{6}$$

$$= \cos\left(2\pi + \frac{\pi}{6}\right)$$

$$= \cos\left(\frac{\pi}{6}\right)$$

Further solving,

$$\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$$

$$= \cos^{-1}\left(\cos\left(\frac{\pi}{6}\right)\right)$$

$$= \frac{\pi}{6}$$

$$\text{Therefore, } \cos^{-1}\left(\cos\frac{13\pi}{6}\right) = \frac{\pi}{6}$$

2. Evaluate $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$

Ans: Consider $\tan\frac{7\pi}{6}$

$$\tan\frac{7\pi}{6}$$

$$=\tan\left(2\pi-\frac{7\pi}{6}\right)$$

$$=\tan\left(\frac{\pi}{6}\right)$$

Further solving,

$$\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$$

$$=\tan^{-1}\left(\tan\left(\frac{\pi}{6}\right)\right)$$

$$=\frac{\pi}{6}$$

Therefore, $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)=\frac{\pi}{6}$

3. Using inverse trigonometric identities, prove that $2\sin^{-1}\frac{3}{5}=\tan^{-1}\frac{24}{7}$

Ans: Assuming $2\sin^{-1}\frac{3}{5}=\alpha$ ----(1)

$$\sin\frac{\alpha}{2}=\frac{3}{5}$$

$$\cos\frac{\alpha}{2}=\sqrt{1-\left(\frac{3}{5}\right)^2}$$

$$\cos\frac{\alpha}{2}=\frac{4}{5}$$

Hence,

$$\sin \alpha = 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right)$$

$$\sin \alpha = \frac{24}{25}$$

$$\cos \alpha = \sqrt{1 - \left(\frac{24}{25} \right)^2}$$

$$\cos \alpha = \frac{7}{25}$$

Therefore,

$$\tan \alpha = \frac{24}{7}$$

$$\alpha = \tan^{-1} \left(\frac{24}{7} \right)$$

From Equation (1), it is proved that

$$2 \sin^{-1} \frac{3}{5} = \tan^{-1} \left(\frac{24}{7} \right)$$

4. Using inverse trigonometric identities, prove that $\sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{77}{36}$

Ans: Assuming $\sin^{-1} \frac{8}{17} = \alpha$

$$\sin \alpha = \frac{8}{17}$$

$$\cos \alpha = \sqrt{1 - \left(\frac{8}{17} \right)^2}$$

$$\cos \alpha = \frac{15}{17}$$

Hence,

Therefore,

$$\tan \alpha = \frac{8}{15}$$

$$\alpha = \tan^{-1} \left(\frac{8}{15} \right)$$

$$\sin^{-1} \left(\frac{8}{17} \right) = \tan^{-1} \left(\frac{8}{15} \right) \text{ ----(1)}$$

Assuming that $\sin^{-1} \frac{3}{5} = \beta$

$$\sin \beta = \frac{3}{5}$$

$$\cos \beta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$\cos \beta = \frac{4}{5}$$

Therefore,

$$\tan \beta = \frac{3}{4}$$

$$\beta = \tan^{-1} \left(\frac{3}{4} \right)$$

$$\sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \text{ ----(2)}$$

Consider

$$\sin^{-1} \left(\frac{8}{17} \right) + \sin^{-1} \left(\frac{3}{5} \right)$$

From Equations (1) and (2),

$$\sin^{-1} \left(\frac{8}{17} \right) + \sin^{-1} \left(\frac{3}{5} \right)$$

$$= \tan^{-1} \left(\frac{8}{15} \right) + \tan^{-1} \left(\frac{3}{4} \right)$$

$$= \tan^{-1} \left(\frac{\frac{8}{15} + \frac{3}{4}}{1 - \left(\frac{8}{15} \times \frac{3}{4} \right)} \right)$$

$$= \tan^{-1} \left(\frac{77}{36} \right)$$

Hence, it is proved that $\sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{77}{36}$

5. Using inverse trigonometric identities, prove that $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$

Ans: Assuming $\cos^{-1} \frac{4}{5} = \alpha$

$$\cos \alpha = \frac{4}{5}$$

$$\sin \alpha = \sqrt{1 - \left(\frac{4}{5}\right)^2}$$

$$\sin \alpha = \frac{3}{5}$$

Therefore,

$$\tan \alpha = \frac{3}{4}$$

$$\alpha = \tan^{-1} \left(\frac{3}{4} \right)$$

$$\cos^{-1} \left(\frac{4}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \text{ ----(1)}$$

Assuming that $\cos^{-1} \frac{12}{13} = \beta$

$$\cos \beta = \frac{12}{13}$$

$$\sin \beta = \sqrt{1 - \left(\frac{12}{13}\right)^2}$$

$$\sin \beta = \frac{5}{13}$$

Therefore,

$$\tan \beta = \frac{5}{12}$$

$$\beta = \tan^{-1} \left(\frac{5}{12} \right)$$

$$\cos^{-1} \left(\frac{12}{13} \right) = \tan^{-1} \left(\frac{5}{12} \right) \text{ ----(2)}$$

Consider

$$\cos^{-1} \left(\frac{4}{5} \right) + \cos^{-1} \left(\frac{12}{13} \right)$$

From Equations (1) and (2),

$$\begin{aligned}
 & \cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) \\
 &= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{5}{12}\right) \\
 &= \tan^{-1}\left(\frac{\frac{3}{4} + \frac{5}{12}}{1 - \left(\frac{3}{4} \times \frac{5}{12}\right)}\right) \\
 &= \tan^{-1}\left(\frac{56}{33}\right) \\
 &= \cos^{-1}\left(\frac{33}{65}\right)
 \end{aligned}$$

Hence, it is proved that $\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13} = \cos^{-1}\frac{33}{65}$

6. Using inverse trigonometric identities, prove that $\cos^{-1}\frac{12}{13} + \sin^{-1}\frac{3}{5} = \sin^{-1}\frac{56}{65}$

Ans: Assuming $\cos^{-1}\frac{12}{13} = \alpha$

$$\cos \alpha = \frac{12}{13}$$

$$\sin \alpha = \sqrt{1 - \left(\frac{12}{13}\right)^2}$$

$$\sin \alpha = \frac{5}{13}$$

Therefore,

$$\tan \alpha = \frac{5}{12}$$

$$\alpha = \tan^{-1}\left(\frac{5}{12}\right)$$

$$\cos^{-1}\left(\frac{12}{13}\right) = \tan^{-1}\left(\frac{5}{12}\right) \text{ ----(1)}$$

Assuming that $\sin^{-1}\frac{3}{5} = \beta$

$$\sin \beta = \frac{3}{5}$$

$$\cos \beta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$\cos \beta = \frac{4}{5}$$

Therefore,

$$\tan \beta = \frac{3}{4}$$

$$\beta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\sin^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{3}{4}\right) \text{ ----(2)}$$

Consider

$$\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right)$$

From Equations (1) and (2),

$$\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right)$$

$$= \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{3}{4}\right)$$

$$= \tan^{-1}\left(\frac{\frac{3}{4} + \frac{5}{12}}{1 - \left(\frac{3}{4} \times \frac{5}{12}\right)}\right)$$

$$= \tan^{-1}\left(\frac{56}{33}\right)$$

$$= \sin^{-1}\left(\frac{56}{65}\right)$$

Hence, it is proved that $\cos^{-1}\frac{12}{13} + \sin^{-1}\frac{3}{5} = \sin^{-1}\frac{56}{65}$

7. Using inverse trigonometric identities, prove that $\tan^{-1}\frac{63}{16} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5}$

Ans: Assuming $\sin^{-1} \frac{5}{13} = \alpha$

$$\sin \alpha = \frac{5}{13}$$

$$\cos \alpha = \sqrt{1 - \left(\frac{5}{13}\right)^2}$$

$$\cos \alpha = \frac{12}{13}$$

Therefore,

$$\tan \alpha = \frac{5}{12}$$

$$\alpha = \tan^{-1} \left(\frac{5}{12} \right)$$

$$\sin^{-1} \left(\frac{5}{13} \right) = \tan^{-1} \left(\frac{5}{12} \right) \text{ ----(1)}$$

Assuming that $\cos^{-1} \frac{3}{5} = \beta$

$$\cos \beta = \frac{3}{5}$$

$$\sin \beta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$\sin \beta = \frac{4}{5}$$

Therefore,

$$\tan \beta = \frac{4}{3}$$

$$\beta = \tan^{-1} \left(\frac{4}{3} \right)$$

$$\cos^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{4}{3} \right) \text{ ----(2)}$$

Consider

$$\sin^{-1} \left(\frac{5}{13} \right) + \cos^{-1} \left(\frac{3}{5} \right)$$

From Equations (1) and (2),

$$\begin{aligned} & \sin^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(\frac{3}{5}\right) \\ &= \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{4}{3}\right) \\ &= \tan^{-1}\left(\frac{\frac{4}{3} + \frac{5}{12}}{1 - \left(\frac{4}{3} \times \frac{5}{12}\right)}\right) \\ &= \tan^{-1}\left(\frac{63}{16}\right) \end{aligned}$$

Hence, it is proved that $\tan^{-1}\frac{63}{16} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5}$

8. Evaluate $\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}$

Ans: We know that $\tan^{-1}a + \tan^{-1}b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$

Consider $\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}$

$$\begin{aligned} & \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8} \\ &= \tan^{-1}\left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \left(\frac{1}{5} \times \frac{1}{7}\right)}\right) + \tan^{-1}\left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \left(\frac{1}{3} \times \frac{1}{8}\right)}\right) \\ &= \tan^{-1}\left(\frac{6}{17}\right) + \tan^{-1}\left(\frac{11}{23}\right) \\ &= \tan^{-1}\left(\frac{325}{325}\right) \\ &= \tan^{-1}(1) \\ &= \frac{\pi}{4} \end{aligned}$$

Therefore, $\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

9. Using inverse trigonometric identities, prove that

$$\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right), x \in [0, 1]$$

Ans: Assuming $x = \tan^2 \alpha$

Consider

$$\begin{aligned} \tan^{-1} \sqrt{x} &= \tan^{-1} \sqrt{\tan^2 \alpha} \\ &= \tan^{-1} \tan \alpha \\ &= \alpha \\ \tan^{-1} \sqrt{x} &= \alpha \text{ -----(1)} \end{aligned}$$

Consider

$$\begin{aligned} \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) &= \frac{1}{2} \cos^{-1} \left(\frac{1-\tan^2 \alpha}{1+\tan^2 \alpha} \right) \\ &= \frac{1}{2} \cos^{-1} \cos 2\alpha \\ &= \alpha \\ \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) &= \alpha \text{ -----(2)} \end{aligned}$$

From Equations (1) and (2),

$$\text{It is proved that } \tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right)$$

10. Using inverse trigonometric identities, prove that

$$\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4} \right)$$

Ans: We know that

$$\sqrt{1+\sin x} = \cos \frac{x}{2} + \sin \frac{x}{2}$$

$$\sqrt{1-\sin x} = \cos \frac{x}{2} - \sin \frac{x}{2}$$

Consider

$$\begin{aligned} & \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) \\ &= \cot^{-1} \left(\frac{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) + \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)}{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) - \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)} \right) \\ &= \cot^{-1} \left(\cot \frac{x}{2} \right) \\ &= \frac{x}{2} \end{aligned}$$

It is proved that $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}$

11. Using inverse trigonometric identities, prove that

$$\tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x, -\frac{1}{\sqrt{2}} \leq x \leq 1$$

Ans: Assuming

$$x = \cos 2\beta$$

$$\sqrt{1+x}$$

$$= \sqrt{1+\cos 2\beta}$$

$$= \sqrt{2} \cos \beta$$

$$\sqrt{1-x}$$

$$= \sqrt{1-\cos 2\beta}$$

$$= \sqrt{2} \sin \beta$$

Consider

$$\begin{aligned}
 & \tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) \\
 &= \tan^{-1} \left(\frac{\sqrt{2}\cos\beta - \sqrt{2}\sin\beta}{\sqrt{2}\cos\beta + \sqrt{2}\sin\beta} \right) \\
 &= \tan^{-1} \left(\frac{1 - \tan\beta}{1 + \tan\beta} \right) \\
 &= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \beta \right) \right) \\
 &= \frac{\pi}{4} - \beta \\
 &= \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x
 \end{aligned}$$

It is proved that $\tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x$

12. Using inverse trigonometric identities, prove that

$$\frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \frac{1}{3} = \frac{9}{4} \sin^{-1} \left(\frac{2\sqrt{2}}{3} \right)$$

Ans: Consider

$$\begin{aligned}
 & \frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \frac{1}{3} \\
 &= \frac{9}{4} \left(\frac{\pi}{2} - \sin^{-1} \frac{1}{3} \right) \\
 &= \frac{9}{4} \cos^{-1} \frac{1}{3}
 \end{aligned}$$

Assuming

$$\cos^{-1} \frac{1}{3} = \alpha$$

$$\cos \alpha = \frac{1}{3}$$

$$\sin \alpha = \sqrt{1 - \left(\frac{1}{3}\right)^2}$$

$$\sin \alpha = \frac{2\sqrt{2}}{3}$$

$$\alpha = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$$

$$\cos^{-1}\frac{1}{3} = \sin^{-1}\frac{2\sqrt{2}}{3}$$

Further solving,

$$\frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3}$$

$$= \frac{9}{4}\cos^{-1}\frac{1}{3}$$

$$= \frac{9}{4}\sin^{-1}\frac{2\sqrt{2}}{3}$$

Therefore, It is proved that $\frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3} = \frac{9}{4}\sin^{-1}\frac{2\sqrt{2}}{3}$

13. For what value of x does the equation $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$ satisfy?

Ans: Consider $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$

$$2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$$

$$\tan^{-1}\left(\frac{2\cos x}{1 - \cos^2 x}\right) = \tan^{-1}(2\operatorname{cosec} x)$$

$$\frac{2\cos x}{1 - \cos^2 x} = 2\operatorname{cosec} x$$

$$\sin x \cos x = \sin^2 x$$

$$\sin x (\cos x - \sin x) = 0$$

$$\sin x = 0, \cos x - \sin x = 0$$

$$x = 0, \frac{\pi}{4}$$

Therefore, $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$ is satisfied for $x = 0, \frac{\pi}{4}$

14. For what value of x does the equation $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x, (x > 0)$ satisfy?

Ans: Consider $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x$

$$\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x$$

$$\tan^{-1} \left(\tan \left(\frac{\pi}{4} - \tan^{-1} x \right) \right) = \frac{1}{2} \tan^{-1} x$$

$$\frac{\pi}{4} - \tan^{-1} x = \frac{1}{2} \tan^{-1} x$$

$$\frac{\pi}{4} = \frac{3}{2} \tan^{-1} x$$

$$\tan^{-1} x = \frac{\pi}{6}$$

$$x = \frac{1}{\sqrt{3}}$$

Therefore, $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x$ is satisfied for $x = \frac{1}{\sqrt{3}}$

15. The expression $\sin(\tan^{-1} x), |x| < 1$ is equal to

(A) $\frac{x}{\sqrt{1-x^2}}$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1+x^2}}$

(D) $\frac{x}{\sqrt{1+x^2}}$

Ans: Assuming

$$x = \tan \beta$$

$$\beta = \tan^{-1} x$$

$$\sin(\tan^{-1} x)$$

$$= \sin \beta$$

$$= \frac{x}{\sqrt{1+x^2}}$$

$$\text{Hence, } \sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$$

Therefore, the correct option is option D

16. For what value of x does the equation $\sin^{-1}(1-x) - 2\sin^{-1} x = \frac{\pi}{2}$ satisfy?

(A) $0, \frac{1}{2}$

(B) $1, \frac{1}{2}$

(C) 0

(D) $\frac{1}{2}$

Ans: Consider

$$\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$$

$$-2\sin^{-1}x = \frac{\pi}{2} - \sin^{-1}(1-x)$$

$$-2\sin^{-1}x = \cos^{-1}(1-x) \text{ -----(1)}$$

Assuming

$$\cos^{-1}(1-x) = \beta$$

$$\cos\beta = 1-x$$

$$\sin\beta = \sqrt{1-(1-x)^2}$$

$$\sin\beta = \sqrt{2x-x^2}$$

$$\beta = \sin^{-1}\sqrt{2x-x^2}$$

$$\cos^{-1}1-x = \sin^{-1}\sqrt{2x-x^2} \text{ -----(2)}$$

Substituting Equation (2) in (1)

$$-2\sin^{-1}x = \sin^{-1}\sqrt{2x-x^2}$$

$$\sin^{-1}(-2x\sqrt{1-x^2}) = \sin^{-1}\sqrt{2x-x^2}$$

$$-2x\sqrt{1-x^2} = \sqrt{2x-x^2}$$

$$4x^2 - 4x^4 = 2x - x^2$$

$$4x^4 - 5x^2 + 2x = 0$$

$$x = 0, \frac{1}{2}, \frac{-1 \pm \sqrt{17}}{4}$$

But considering the given options and also when $x = \frac{1}{2}$, the equation

$$\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2} \text{ doesn't satisfy}$$

Thus, $x=0$ is the only solution.

Hence the correct option is C

17. Solving $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\frac{x-y}{x+y}$ results in

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $-\frac{3\pi}{4}$

Ans: Consider

$$\begin{aligned} & \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\frac{x-y}{x+y} \\ &= \tan^{-1}\left(\frac{\frac{x}{y} - \frac{x-y}{x+y}}{1 + \left(\frac{x}{y} \times \frac{x-y}{x+y}\right)}\right) \\ &= \tan^{-1}\left(\frac{x^2+y^2}{y^2+x^2}\right) \\ &= \tan^{-1}1 \\ &= \frac{\pi}{4} \end{aligned}$$

Therefore, $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\frac{x-y}{x+y} = \frac{\pi}{4}$

Hence, the correct option is option C